How accurately can the vest tell where a sound came from?

Eight ways to place the 8 mics, compared on two jobs: knowing where a sudden sound like an explosion came from, and picking out the wearer's voice from factory noise. Every number on this page is computed from the mic positions.

Best for both: collar front + shoulders + upper back

Four mics on the front of the collar, one on each shoulder, two on the upper back. It finds a sharp bang to about ±1.6° and a distant blast to about ±3.7°, all the way round with no front/back mix-ups. It also has the best voice pickup of the eight (+7.7 dB over one chest mic), and its back pair, shielded by the body, serves as the noise reference. The planned vest locates a little better but hears the voice about 3 dB worse. Chest-only layouts can't tell front from back, so they fail at "where did that come from".

Sound: Sharp bang (impact, dropped tool, balloon pop) | Distant blast (low boom, mostly below 1 kHz)

☑ Fabric moves the mics about ±5 mm ☐ Echoey steel building

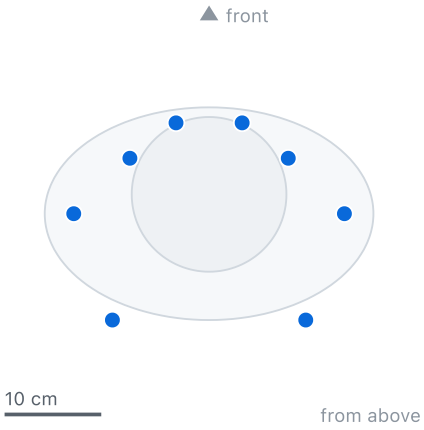
All eight layouts: distant blast

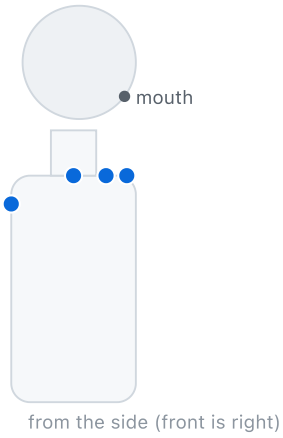
Layout	Direction error (typical)	Worst direction	Off by, at 10 m	Front/back mix-ups	Voice vs noise	Noise reference mics
Line across the chest	±9.9°	blind	±1.7 m	100% of directions	+2.1 dB	none
9 cm ring flat on the chest	±25°	blind	±4.7 m	100% of directions	+3.2 dB	none
9 cm ring on top of the helmet	±8.7°	±8.7°	±1.5 m	none	+0.4 dB	none
Collar ring around the neck	±4.6°	±5.0°	±0.8 m	none	+7.4 dB	none
Two 4-mic rings, one per shoulder	±2.9°	±14°	±0.5 m	11% of directions	+5.3 dB	none
Vest layout (planned)	±3.3°	±4.8°	±0.6 m	none	+4.9 dB	2 (voice -13 dB)
Vest, chest line + shoulders + back	±3.7°	±4.9°	±0.7 m	none	+3.9 dB	2 (voice -13 dB)
Collar front + shoulders + upper back **(recommended)**	±3.7°	±6.0°	±0.7 m	none	+7.7 dB	2 (voice -14 dB)

Direction error: 1-sigma error of the bearing, median over all 360° (the best a good estimator such as SRP-PHAT can reach). Off by at 10 m: how far sideways that bearing misses a blast 10 m away. Voice vs noise: a beam aimed at the mouth compared with one chest mic 20 cm from the mouth, with noise from all around, 300–3400 Hz. Noise references: mics that hear at least 10 dB less voice, for an adaptive canceller.

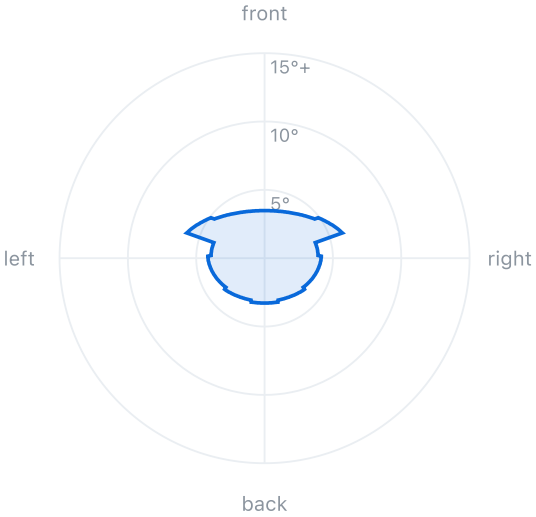
Line across the chest | 9 cm ring flat on the chest | 9 cm ring on top of the helmet | Collar ring around the neck | Two 4-mic rings, one per shoulder | Vest layout (planned) | Vest, chest line + shoulders + back | Collar front + shoulders + upper back

Where the mics go





Direction error by where the sound comes from



Radius is the 1-sigma bearing error in degrees: closer to the centre is better, and anything past 15° is drawn on the outer ring. An orange band outside the ring marks directions the array can confuse with another direction (usually its front/back mirror).

Collar front + shoulders + upper back

±3.7°	+7.7 dB	14 cm
typical bearing error	voice vs noise	nearest mic to mouth

Good
Best of both. Four front-collar mics 14 cm from the mouth for voice; shoulders and upper back spread the array for 360°; the back pair is shielded by the body and works as noise references.

Watch out
The collar mics need a stiff collar or a sewn-in strip so they don't flop around. Keep them clear of the radio's own mic.

Wiring: the four collar mics on D0 and D3, shoulders on D4, back pair on D5. Same board and clock as today.

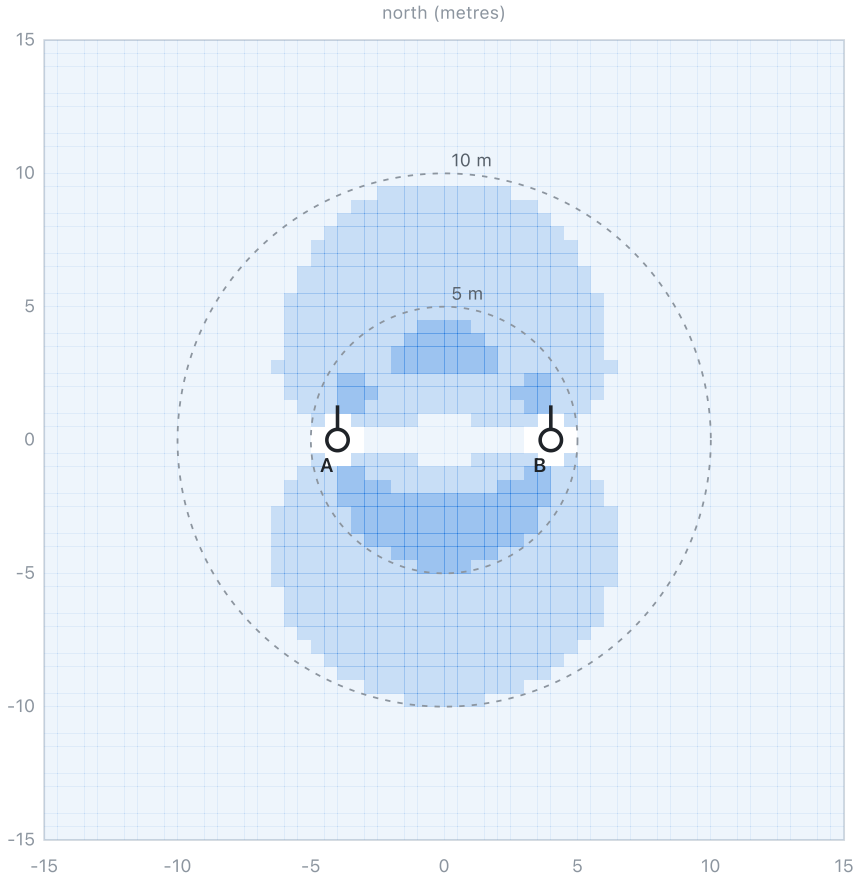
Locating the blast with several vests

One vest gives a direction, not a distance. When two or three workers' vests hear the same blast, their bearings cross at the source. The map shows how far off that crossing point would be for a distant blast anywhere on a 30 × 30 m floor, with every worker wearing the **collar front + shoulders + upper back** layout.

Workers: A + B | A + B + C

Each vest knows which way it faces to: perfect | ±2° | ±5° | ±10°

Worker position known to: exact | ±0.5 m | ±1 m



under 0.3 m | 0.3–0.6 m | 0.6–1.2 m | 1.2–2.5 m | over 2.5 m | front/back mix-up possible

Position error where the bearings cross (RMS, metres) for a blast at each 0.5 m square. Workers A and B stand 8 m apart facing north; C stands 7 m behind them facing east. Lines from each worker show which way they face. Squares within 1 m of a worker are left blank.

2.2 m
typical error within 10 m

1.1 m
if heading and position were perfect

What limits the map

With good layouts the mics are no longer the weak link. A ±5° error in knowing which way the worker faces costs far more than the 2–4° bearing error of the array, so the vest needs a decent heading sensor (IMU with gyro, calibrated away from steel) and a position for each worker (UWB or BLE beacons) before multi-vest location is worth showing.

Error grows with distance and along the line through two workers, where their bearings are nearly parallel. A third worker off that line fixes most of it.

Bearings only need each vest's own clock, so the vests don't have to be synchronised; they just send "blast at 40° right, ±4°" plus their heading over the radio.

Explosions specifically

It will clip. Typical MEMS mics max out around 120 dB SPL, and a nearby blast is louder. A clipped waveform still has a sharp start, so time the onset (the first few milliseconds), not the whole waveform. All eight mics clipping at once is itself a strong explosion trigger.

A low boom is harder to time. That is the "distant blast" setting: about four times the timing error of a sharp bang, so roughly twice the bearing error once fabric movement is included.

Echoes come later. In a steel building the first arrival is the true direction; reflections follow within milliseconds. Use only the onset window, and tick "Echoey steel building" above to see the cost.

Keep a raw path. The detector should listen to the unprocessed mics, never the noise-cancelled voice channel, so blasts and alarms are never filtered out.

How to check these numbers

Test	What to compare
Balloon pops at 2 m from 8 directions around the mannequin	Measured bearing error against the sharp-bang column
A heavy board dropped flat on the floor (low thump)	Against the distant-blast column
Same with factory noise from a speaker	How fast the error grows with noise
Talk with noise playing; run noise_compare.py	Voice vs noise gain between layouts

Model assumptions: each mic's arrival time is known to 10 µs for a sharp bang or 40 µs for a low boom, plus 15 µs for fabric movement and 20 µs for echoes when ticked. Mics facing away from a sound get 8 dB less (torso) or 4 dB less (neck, helmet, shoulder top). Sounds arrive level, at ear height. The mouth is 14 cm above and 9 cm in front of the shoulder line. Real results will be worse than this best case; the ranking between layouts is the useful part.